Chapter 1 — Measurements

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Physical quantities and SI units

Q.1 Explain the difference between base and derived units and state the SI conventions for writing units.

Base units are the seven independent units of the International System: metre (m) for length, kilogram (kg) for mass, second (s) for time, ampere (A) for electric current, kelvin (K) for temperature, mole (mol) for amount of substance and candela (cd) for luminous intensity. Derived units are obtained by multiplying or dividing base units, for example newton N = kg m s^{-2}, joule J = N m, watt W = J s^{-1}, pascal Pa = N m^{-2}, coulomb C = A s and volt V = J C^{-1}.

SI conventions make scientific writing uniform. Unit symbols are not pluralised and are not followed by a full stop; a space is left between the number and the unit; prefixes such as kilo, milli and micro are attached directly to the unit. Compound prefixes are not allowed, so 1 μμF must be written as 1 pF. When a multiple is raised to a power, the power applies to the whole multiple: 1 cm^3 = $(10^{-2}$ m$)^3$ = 10^{-6} m^3.

Base quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Important Conceptual Short Questions — Topic 1

1. Why is a common system of units necessary in physics?

A common system allows measurements made in different laboratories and countries to be compared without confusion. SI units give a single standard for every physical quantity, so equations and experimental results have the same meaning everywhere.

2. Why are compound prefixes not allowed in SI?

Compound prefixes such as μμ make the order of magnitude unclear and increase the chance of error. SI allows only one prefix per unit, so 1 μμF is correctly written as 1 pF.

3. How does a power apply to a prefixed unit?

The power acts on the whole prefixed unit, not only on the base unit. For example, 1 cm^3 means $(10^{-2}\ m)^3 = 10^{-6}\ m^3$, not $10^{-2}\ m^3$.

4. How can a reasonable estimate of a physical quantity be made?

Compare the quantity with a familiar standard and keep only the order of magnitude. For example, the mass of a student can be estimated as about 50 kg by comparison with known masses, which is useful for checking whether a calculated answer is sensible.

Topic 2 — Uncertainty in measurement

Q.2 Justify why every measurement contains uncertainty and explain absolute, fractional and percentage uncertainty.

No instrument can read a quantity exactly because it has a finite least count and because random errors affect repeated readings. The uncertainty is therefore an estimate of the range within which the true value is expected to lie. For a single reading it is usually taken as the least count of the instrument; for repeated readings the mean deviation from the average is used.

If a length is recorded as $x = 25.5 \pm 0.1$ cm, the absolute uncertainty is $\Delta x = 0.1$ cm. The fractional uncertainty is $\Delta x/x = 0.1/25.5 \approx 0.0039$, and the percentage uncertainty is $0.0039 \times 100 \approx 0.39\%$. Absolute uncertainty tells the precision of the instrument, while fractional or percentage uncertainty tells how accurate the measurement is relative to its size.

Textbook check (p. 10): the “Travel time of light” box prints Moon to Earth as 1 min 20 s. Light takes about 1.3 s from the Moon to the Earth; 8 min 20 s is the Sun–Earth light time.

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Important Conceptual Short Questions — Topic 2

1. Why do repeated readings reduce random uncertainty?

Random errors are equally likely to be positive or negative. Taking the average of many readings cancels part of these errors, and the mean deviation gives a smaller estimate of the uncertainty than a single reading.

2. How does the least count of an instrument limit a measurement?

The least count is the smallest change the instrument can show. A reading cannot be reliably quoted to a smaller division than this, so the least count fixes the smallest possible absolute uncertainty of a single reading.

3. Why does the last digit of a digital scale sometimes fluctuate?

The last digit corresponds to the smallest step of the digital scale. Small random changes in the measured quantity or in the electronics can move the reading by one or two steps, so that digit is uncertain and should still be recorded.

Topic 3 — Significant figures

Q.3 Describe the rules for counting significant figures and rounding, and explain how they are used in calculations.

All non-zero digits are significant. Zeros between non-zero digits are significant; leading zeros are not significant; trailing zeros after a decimal point are significant. Trailing zeros in a whole number are ambiguous unless a decimal point or scientific notation is used. In rounding, if the dropped digit is greater than 5 the last kept digit is increased, and if it is less than 5 it is left unchanged. If the dropped part is exactly 5 followed by zeros, the book uses the even-digit rule: the last kept digit is made even, so 43.75 rounds to 43.8 and 73.650 rounds to 73.6.

In multiplication and division the answer is given to the same number of significant figures as the least precise factor. In addition and subtraction the answer is given to the same number of decimal places as the term with the fewest decimal places. Thus $2.8723 \times 1.6 = 4.6$ (2 significant figures), while 12 kg + 2.02 kg + 5.1 kg = 19 kg (no decimal place).

Important Conceptual Short Questions — Topic 3

1. Why are leading zeros not counted as significant figures?

Leading zeros only locate the decimal point; they do not come from the measurement. For example, 0.0025 has two significant figures, 2 and 5, because the zeros only show the size of the number.

2. Why is the even-digit rule used when the dropped digit is exactly 5?

Rounding 5 always upward would bias many results high. Making the last kept digit even rounds half the cases up and half down, so the average rounding error is reduced.

3. How does scientific notation remove ambiguity in significant figures?

Scientific notation shows exactly which digits are measured. For example, 2.50×10^3 clearly has three significant figures, whereas 2500 may have two, three or four.

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Topic 4 — Precision and accuracy

Q.4 Differentiate between precision and accuracy and relate them to absolute and fractional uncertainty.

Precision means how close repeated measurements are to one another. It depends on the least count of the instrument and is expressed by absolute uncertainty. A vernier callipers with least count 0.01 cm is more precise than a metre rule with least count 0.1 cm. Accuracy means how close a measurement is to the true value. It is expressed by fractional or percentage uncertainty; the smaller the percentage uncertainty, the more accurate the measurement.

Quantity	Precision	Accuracy
Meaning	Closeness of repeated readings to each other	Closeness of a reading to the true value
Depends on	Least count of the instrument	Fractional or percentage uncertainty
Uncertainty used	Absolute uncertainty	Fractional / percentage uncertainty
Improved by	Using a finer instrument	Reducing percentage uncertainty and removing systematic error

Important Conceptual Short Questions — Topic 4

1. Why can a measurement be precise but not accurate?

A precise instrument gives closely agreeing readings, but a zero error or wrong calibration can shift all readings away from the true value. The readings are then consistent with each other but not correct.

2. How does a zero error affect accuracy?

A zero error adds the same offset to every reading, so it is a systematic error. It does not reduce precision, but it makes the measurements inaccurate until the offset is corrected.

3. Why does a smaller least count give greater precision?

A smaller least count allows smaller differences to be detected, so repeated readings can agree more closely. This reduces the absolute uncertainty and increases precision.

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Topic 5 — Assessment of total uncertainty

Q.5 Derive the rules for combining uncertainties in sums, products and powers.

For addition or subtraction, absolute uncertainties are added. If $x = x_1 \pm x_2$, then $\Delta x = \Delta x_1 + \Delta x_2$. For example, 15.4 ± 0.1 cm and 25.6 ± 0.1 cm give a difference of 10.2 ± 0.2 cm. For multiplication or division, percentage uncertainties are added. If R = V/I, V = 3.4 ± 0.1 V (3%) and I = 0.68 ± 0.05 A (7%), then R = 5.0 Ω with 10% uncertainty, so R = 5.0 ± 0.5 Ω.

For a power, the percentage uncertainty is multiplied by the power factor. If $A = \pi r^2$ and r = 1.25 ± 0.01 cm, the percentage uncertainty in r is 0.8%, so the percentage uncertainty in A is 2 × 0.8% = 1.6%. These rules give the maximum possible uncertainty in a derived quantity.

Important Conceptual Short Questions — Topic 5

1. Why are absolute uncertainties added even when quantities are subtracted?

Uncertainty is a range, not a signed error. In the worst case the errors of the two quantities act in opposite directions, so the possible spread of the result is the sum of the two absolute uncertainties.

2. Why are percentage uncertainties added in multiplication and division?

A fractional error in each factor contributes proportionally to the fractional error of the product or quotient. Adding the percentage uncertainties gives the maximum possible relative error of the result.

3. How does a power increase the uncertainty of a derived quantity?

If a quantity is raised to the power n, its fractional uncertainty is multiplied by n. For example, area depends on r^2, so the percentage uncertainty in area is twice that in radius.

Topic 6 — Dimensions of physical quantities

Q.6 Explain the dimensions of physical quantities and describe the uses and limitations of dimensional analysis.

The dimensions of a quantity show how it depends on the base quantities mass [M], length [L] and time [T]. Force has dimensions $[MLT^{-2}]$, energy $[ML^2T^{-2}]$, power $[ML^2T^{-3}]$, pressure $[ML^{-1}T^{-2}]$, Planck's constant $[ML^2T^{-1}]$ and angular velocity $[T^{-1}]$. Dimensional analysis is used to check the correctness of equations, to derive simple relations and to convert units.

Its limitations are important. It cannot determine dimensionless constants such as ½ or 2π; it cannot handle trigonometric, exponential or logarithmic functions; and it cannot distinguish quantities with the same dimensions, such as work and torque. A dimensionally correct equation may still be physically wrong.

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Important Conceptual Short Questions — Topic 6

1. Why can a dimensionally correct equation still be wrong?

Dimensions only check the type of quantity on each side. They cannot check numerical factors, dimensionless constants or the actual physical law, so an equation can pass the dimensional test and still be incorrect.

2. How can dimensions be used to check an equation?

Replace each quantity by its dimensions and simplify both sides. If the dimensions are identical, the equation is dimensionally consistent; if they differ, the equation must be wrong.

3. Why is angle considered dimensionless?

Angle in radians is the ratio of arc length to radius, [L]/[L] = 1. It has no base dimensions, although the unit radian is still written for clarity.